

KUNYUAN HU

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Education

University of Toronto Mississauga

Expected May 2029

Honours Bachelor of Science, Computer Science and Statistics (Double Major)

Mississauga, ON

- **Coursework:** Software Design, Data Structures & Analysis, Introduction to Computer Science, Probability & Statistics I, Linear Algebra, Calculus
- **GPA:** 3.42 / 4.00

Projects

ClipFarm — Automated Volleyball Highlight Generator

Mar 2026 – Present

Python, FastAPI, Next.js, TypeScript, YOLOv8-pose, Roboflow, PostgreSQL, Docker, Celery

- Built the rally-detection stage of a highlight pipeline: a custom-trained **Roboflow** ball detector bounds rallies by trajectory, and **audio cheer detection** plus rally-shape scoring rank which rallies become highlights.
- Classified player actions with **YOLOv8-pose** and cut final clips with **FFmpeg**, offloading GPU inference to **Modal** with content-hash caching so repeat uploads of the same footage skip inference entirely.
- Contributed to the **Next.js + TypeScript** frontend — filterable clip grid and lazy-loaded thumbnails — served by **FastAPI** with **Celery + Redis** job distribution over **PostgreSQL**, containerized with **Docker Compose**.

Imposter — Real-Time Multiplayer Party Game

2026

React, Node.js, WebSockets, Vite

- Built a real-time game that keeps player phones in sync through timed clue and voting phases, using a **Node WebSocket server** as the authoritative source of room state.
- Implemented **four-letter room codes**, turn sequencing, and phase transitions, handling reconnects and mid-game joins without desyncing clients.

Medi-Cal Scheduling — Surgeon-Patient Booking System

2024

Python, Django, SQLite

- Built a full-stack **Django** appointment booking application organized around a patient-facing calendar rather than a form queue.
- Implemented personal calendar views, booking, and cancellation flows backed by a relational schema in **SQLite**.

Worst Birthday UI — Adversarial Interface Study

2024

Python, Pygame

- Designed a deliberately hostile date-entry interface — blackjack and a crippled calculator stand between the user and their birthday — as a study of **interaction cost** and what makes conventional UI patterns work.

Experience

Server, Part-Time

Dec 2022 – Mar 2024

The Bradley Gracious Retirement Living

Mississauga, ON

- Served residents daily in an assisted living community, balancing speed of service against individual dietary needs and accessibility requirements.
- Held the role for **15 months** alongside full-time studies, coordinating with kitchen and care staff through peak service.

Technical Skills

- **Languages:** Python, TypeScript, JavaScript, Java, SQL, HTML/CSS
- **Frameworks:** React, Next.js, Node.js, Django, FastAPI, Celery, Pygame
- **ML & Data:** YOLOv8-pose, Roboflow, OpenCV, PostgreSQL, SQLite
- **Tools:** Git, Docker, FFmpeg, WebSockets, Redis, Modal, Vite, REST APIs
- **Spoken:** English, French (**DELF B1**), Mandarin